

A Weak-Net Dini Failure For A Non-Normal Cone On ℓ^1

Automatic Math Discovery Run `fa.banach.001`

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Source And Target

Jochen Glueck, *Increasing sequences in ordered Banach spaces – new theorems and open problems*, arXiv:2409.19768.

The paper gives an explicit non-normal cone on ℓ^1 , notes that the Schur property only controls sequences, and says it is unclear whether every increasing weakly convergent net for that cone is norm convergent. It then poses the broader Dini-normal open problem.

non-normal cone in ℓ^+ .

Example 6.1. Let the underlying index set of the real Banach space ℓ^1 be $\mathbb{N}_0 := \{0, 1, 2, \dots\}$. The set

$$\ell_+^1 := \left\{ x \in \ell^1 \mid x_0 \geq \sum_{k=1}^{\infty} \frac{|x_k|}{2^k} \right\}$$

is a closed cone in ℓ^1 which is not normal, but which has non-empty interior and is thus generating.

Proof. It is straightforward to check that the set ℓ_+^1 is a closed cone. The 0-th canonical unit vector e_0 is an interior point of ℓ_+^1 . Hence, ℓ_+^1 has non-empty interior and is thus, in particular, generating.

To see that ℓ_+^1 is not normal, let $e_k \in \ell^1$ denote the k -th canonical unit vector for each $k \in \mathbb{N}_0$. For every $k \geq 1$ one then has $e_0 \geq 2^k e_k$ with respect to the order induced by ℓ_+^1 , which shows the non-normality. $\square$

Note that the Schur property of ℓ^1 is a condition for sequences only – there exist weakly convergent nets in ℓ^1 that are not norm convergent. It is not clear – at least not to the author – whether in the space ℓ^1 endowed with the cone from Example 6.1 every increasing weakly convergent net is norm convergent. This raises the question whether the preceding example is only an artefact of the fact that one considers sequences instead of nets or whether, on the other hand, the validity of Dini's theorems for all nets implies that the cone is normal. Let us pose this as an open problem:

Open Problem 6.2. Let X be an ordered Banach space and assume that every weakly convergent increasing net is norm convergent. Does it follow that X_+ is normal?

Let us also note that, since a weakly convergent net need not be eventually norm bounded, the answer to Open Problem 6.2 might change if one only considers weakly convergent increasing nets that are, in addition, norm bounded.

Finally, we give a partial answer to Open Problem 6.2 by showing that, in a large class of spaces with non-normal cone, Dini's theorem does not hold – not even for sequences.

Theorem 6.3. *Let X be an ordered Banach space with non-normal cone. Assume that the space X is reflexive or that, more generally, the bidual space X'' is a space*

Result

Theorem 1. *Let $X = \ell^1(\mathbb{N}_0)$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, and write $x = (x_0, u)$ with $u = (x_k)_{k \geq 1} \in \ell^1(\mathbb{N})$. Endow X with the cone*

$$X_+ = \left\{ x \in \ell^1(\mathbb{N}_0) : x_0 \geq \sum_{k=1}^{\infty} \frac{|x_k|}{2^k} \right\}.$$

Then there is an increasing net (y_i) in X_+ which converges weakly in X but does not converge in norm.

Proof

For $u = (u_k)_{k \geq 1} \in \ell^1(\mathbb{N})$, put

$$\|u\|_w = \sum_{k=1}^{\infty} \frac{|u_k|}{2^k}.$$

Thus $(a, u) \in X_+$ if and only if $a \geq \|u\|_w$, and

$$(a, u) \leq (b, v) \iff b - a \geq \|v - u\|_w.$$

Lemma 1. *Let $F \subset \ell^\infty(\mathbb{N})$ be finite and let $n \geq 1$. There is $u = u_{n,F} \in \ell^1(\mathbb{N})$ such that*

$$\|u\|_1 = 1, \quad \text{supp } u \subseteq \{k \geq n\}, \quad \varphi(u) = 0 \quad (\varphi \in F).$$

Consequently $\|u\|_w \leq 2^{-n}$.

Proof. Choose $r > |F|$ distinct integers $k_1, \dots, k_r \geq n$. The homogeneous system

$$\sum_{j=1}^r c_j \varphi(k_j) = 0 \quad (\varphi \in F)$$

has a nonzero real solution $c = (c_1, \dots, c_r)$. Define u by supporting it on the k_j 's with coefficients c_j , then normalize so that $\|u\|_1 = 1$. Since $2^{-k_j} \leq 2^{-n}$, we have $\|u\|_w \leq 2^{-n} \|u\|_1 = 2^{-n}$. $\square$

Let D be the set of pairs $i = (n, F)$, where $n \geq 1$ and $F \subset \ell^\infty(\mathbb{N})$ is finite. Define $i = (n, F) \preceq j = (m, G)$ if either $i = j$, or $m \geq n + 2$ and $F \subseteq G$. This is a directed order: given two indices, take the union of the finite sets and a large enough integer.

For each $i = (n, F)$, choose $u_i = u_{n,F}$ as in Lemma 1, and set

$$a_n = 4 - 2^{1-n}, \quad y_i = (a_n, u_i) \in X.$$

Since $a_n \geq 2$ and $\|u_i\|_w \leq 2^{-n} \leq 1/2$, each y_i is in X_+ .

We claim that (y_i) is increasing. If $i = (n, F) \prec j = (m, G)$, then $m \geq n + 2$. Hence

$$\|u_j - u_i\|_w \leq \|u_i\|_w + \|u_j\|_w \leq 2^{-n} + 2^{-m},$$

while

$$a_m - a_n = 2(2^{-n} - 2^{-m}).$$

Since $m \geq n + 2$, one has $2^{-m} \leq 2^{-n}/4$, and therefore

$$2(2^{-n} - 2^{-m}) \geq 2^{-n} + 2^{-m}.$$

Thus $y_j - y_i \in X_+$, proving monotonicity.

We next prove weak convergence. Let $\psi \in X' = \ell^\infty(\mathbb{N}_0)$. Write $\psi = (\beta, \varphi)$, where $\beta \in \mathbb{R}$ acts on the zeroth coordinate and $\varphi \in \ell^\infty(\mathbb{N})$ acts on the tail. For all sufficiently large indices in the directed order, the finite set component contains φ , so $\varphi(u_i) = 0$. Also $a_n \rightarrow 4$ along the directed set. Hence

$$\psi(y_i) = \beta a_n + \varphi(u_i) \longrightarrow 4\beta.$$

This is exactly weak convergence to $(4, 0) \in X$.

Finally,

$$\|y_i - (4, 0)\|_1 = |a_n - 4| + \|u_i\|_1 = 2^{1-n} + 1$$

for every $i = (n, F)$. The net therefore does not converge to $(4, 0)$ in norm. Since the weak limit is unique, it does not converge in norm at all. $\square$

Scope And Limitations

This answers the concrete uncertainty about the explicit cone in Example 6.1 of the source paper: the Schur property does not rescue Dini’s theorem for weakly convergent nets in that ordered ℓ^1 space.

It is not a full solution of Open Problem 6.2. The construction uses the special tail geometry of the cone

$$x_0 \geq \sum_{k \geq 1} |x_k| 2^{-k}$$

and the ability to annihilate finitely many ℓ^∞ functionals on arbitrarily far tails.

Novelty And Duplicate Check

Local duplicate check: the run registry, solution index, and the prior archived attempt note for `2409.19768_order_sup_normality_probe` were checked. The prior attempt did not promote a result and addressed a different route, namely whether supremum existence of increasing order-bounded sequences might force normality.

No external literature search was performed for this partial packet. Novelty confidence is therefore limited to the local run context and the source paper’s own statement that this concrete net question was unclear to the author.

Verification Notes

The proof has two points worth checking:

1. the finite-annihilation lemma, which is just a homogeneous linear system with more variables than equations;
2. the scalar inequality

$$2(2^{-n} - 2^{-m}) \geq 2^{-n} + 2^{-m} \quad (m \geq n + 2),$$

which is also checked in `code/verify_scalar_inequality.py`.